

Evolutionary Robotics

The curious case of Dr Kammerer

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The curious case of Dr Kammerer

This is a story about

- a curious “cold case”
 - How a scientific debate about evolution became a **matter of live and death**.
 - “Science is done by humans.”
(Werner Heisenberg)
and they struggle, fight, and fail.
 - Sometimes tiny stupid details of a seemingly irrelevant study can be the **tipping point** for or against an essential scientific theory.
- for all the details:
Arthur Koestler,
The Case of the Midwife Toad (English),
or *Der Krötenküsser* (German),
1971



Let me take you please to **Austria in the 1920s.**

Early afternoon of September 23, 1926

A trail on the **Schneeberg**,
with 2,076 m the highest mountain of
Lower Austria

While repairing the hiking trail a worker
finds the **body** of a well dressed man

He sits on the ground
with his back at a rock,
his right hand still holding a gun
that was obviously used
to **shoot in his head**

The worker finds a **suicide note**...

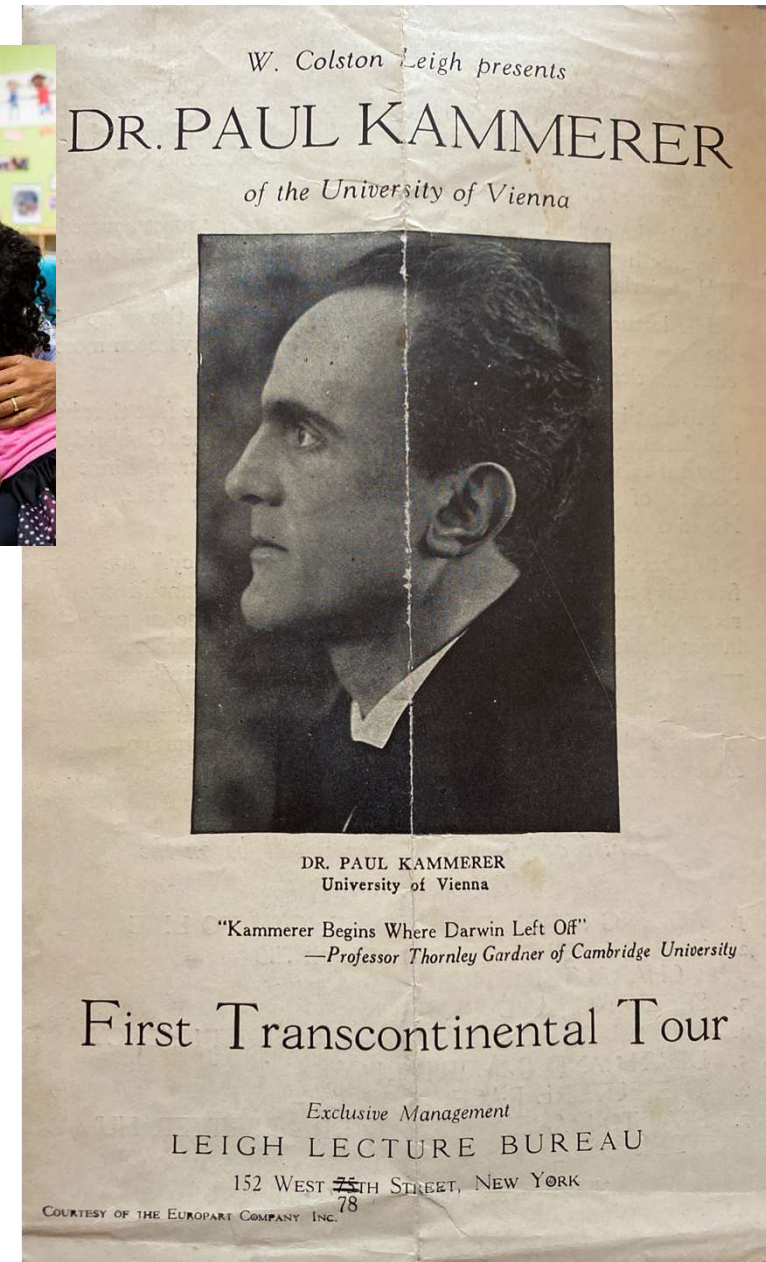


Who was this poor guy
Dr Paul Kammerer?

Dr. Paul Kammerer requests not to be transported to his home, in order to spare his family the sight. Simplest and cheapest would perhaps be utilisation in the dissecting room of one of the university institutes. I would actually prefer to render science at least this small service. Perhaps my worthy academic colleagues will discover in my brain a trace of the qualities they found absent from the manifestations of my mental activities while I was alive. Whatever happens to the corpse, buried, burned, or dissected, its bearer belonged to no religious community and wishes to be spared a religious ceremony, which probably would be denied him anyway. This is not meant to express hostility against the individual priest, who is human like everybody else and often a good and noble person.¹

Kammerer was a star

- He toured the US in 1924/25.
- “Kammerer begins where Darwin left off”
- The New York Times praised him as “next Darwin”
- Seemingly people loved the idea that good behaviors and a positive way of life are a “scientifically proven” way of supporting your children as the idea was they would inherit that.
- This is in contrast to Darwinism where everything you can do in life is “too late” as your genes are already determined.
(later more)



IN THIS SECTION

FIND MISSING YOUTH ON GALLOPS
A REMARKABLE TRAGEDY IN SCIENCE
HIKERS FIND MICHAELMAS DAISES

The World

SECOND NEWS SECTION

NEW YORK, SUNDAY, OCTOBER 17, 1926.

IN THIS SECTION

CHILD OF FORMER ORGAN-PUMPERS
GIRL PSYCHIC POSSESSED BY DEVIL
ACCIDENT SPOOLS A "PERFECT CRIME"

Vienna Scientist Exposed by New Yorker Commits Suicide

Dr. Paul Kammerer's Midwife Toad, or Frog Found, By
Dr. G. K. Noble, of Natural History Museum,
To Have Been Doctored with India Ink

By Ralph P. Taylor

His work was discussed throughout Europe and America. A number of other scientists became converts to his theories. He was hailed by them as "a second Darwin."

Then a very small thing put an end to his career. It was the thumb of a toad he had used to prove his conclusions. It was examined under a microscope and the fragile outer layer of skin was lifted carefully off. Under it was found something which caused the scientist to put a bullet through his brain.

Such, in brief, was the career of Dr. Paul Kammerer, Viennese biologist and philosopher, sometime associate of Dr. Eugene Steinach, discoverer of the Steinach method of rejuvenation by means of surgery.

"When Kammerer first claimed the attention of the scientific world and of the public it was as the disciple and preacher of certain ideas which conservative scientists refused to accept. He possessed theories which smacked of superstition, and he made claims which sounded extremely extravagant."

He had written a book called "The Law of Series," in which he had presented material to show that what we regard as curious coincidences are but the workings of a law which causes events of a particular character to occur in series.

He claimed to have conducted remarkable experiments upon small animals and to have obtained numerous results proving that acquired characteristics may be inherited. He asserted he had caused eyes to grow in blind rats and to be inherited by the offspring of those rats. He claimed he had changed land creatures into water-loving creatures, that he had changed the color of salamanders and the form of mollusks.

He had done this, he said, by keeping some in enclosures in which there was no water and in which they were forced to hatch their eggs in the air, and by keeping others in tanks in which the only land available consisted of small islands and in which the temperature was maintained at a point calculated to cause the creatures to seek the water to cool off.

As a result of these experiments, Kammerer said, the male midwife toads that were compelled to exist altogether on land eventually lost the egg-carrying habit common to their kind, and the males that were forced to live in the water developed after a few generations—what are known as nuptial pads; that is, little horny pads on the thumbs which are common among water frogs and which are used by those creatures to hold on to their mates in the water.

These and other claims made by Kammerer precipitated a tempest of controversy. Scientists here and abroad expressed doubts as to Kammerer's ability to prove his claims. They said others had experimented in the field of acquired characteristics without being able to get any such evidence as Kammerer claimed to support the theory that such characteristics are inherited.

Then Dr. William Bateson, one of the leading biologists of England, made a bitter attack upon the Austrian scientist. Dr. Bateson said that a supposed nuptial pad appearing in a photograph of one specimen was in the wrong place—that is, the pad was so located that it would not help the male in grasping the female. Dr. Bateson's views were expressed in a letter published in 1912 in Nature, a scientific weekly published in London.

But this attack served merely to lead to the water for hatching and others to lose their desire for the land and to become thoroughly aquatic in their habits.

Likewise, Dr. Timothy Glicker, Professor of Zoology at Cambridge.



DR. PAUL KAMMERER, VIENNESE BIOLOGIST AND PHILOSOPHER, SOMETIME ASSOCIATE OF DR. EUGENE STEINACH, DISCOVERER OF THE STEINACH METHOD OF REJUVENATION BY MEANS OF SURGERY.



THE MALE MIDWIFE-FROG FROM THE ROYAL NATURAL HISTORY MUSEUM, LONDON.

Kammerer's Suicide

Letter to Moscow

Kammerer's farewell letter was addressed to the Communist Academy, Moscow. It reads as follows: "Dear Comrades and Colleagues: "You undoubtedly know that my attack in London was of a nature directed against me. The attacks based on Noble's test examination of my experiments on a specimen of albatross with putting weals was made at the Vienna Biological Institute. The principal feature of the accusation in an artificial coloring probably effected with China ink whereby an alleged false impression of black skin discoloration in vicinity of the weals is brought about. This implies an intention for freedom of which I must myself alone will be accused. "After reading the attack I found once more over the text specimen and found Noble's statements entirely confirmed. I found also some of my salamanders having



DR. G. K. NOBLE

to use those hands; how Negroes might become white after generations of unfamiliarity with the tropical sun; how women of fifty might make themselves look like thirty-five; how future generations might arrive with ease at conclusions which the wisest men in the present generation must struggle to attain; how eventually

late, Dr. Hans Fehren of Vienna collaborated in the examination. Dr. Noble was permitted to bring part of the specimen back to New York for minute study.

On Aug. 1, "Nature," the British scientific weekly which carried Bateson's original attack upon Kammerer, carried Noble's statement of the result of his examination of that single specimen on which Kammerer staked his claim to rejuvenation.

The question which Noble wished to settle by his examination was whether or not this toad possessed on its thumbs the so-called nuptial pads common to water frogs. These are small, blackish pads with rough surfaces used for holding the female. If this toad actually had such pads it would appear to support Kammerer's claims; if it did not possess them Kammerer's claims had nothing to uphold them save his own word and some photographic evidence of questionable value.

Noble told in Nature how he had

- news in New York, October 17, 1926
- midwife toad?
- doctored with India ink?
- Dr G. K. Noble?

Then Dr. Noble described a series

Dr Gladwyn Kingsley Noble

The scandal which shook the scientific world and destroyed Kammerer's reputation exploded in the columns of *Nature* on August 7, 1926. On that day the journal published an article by Dr. G. K. Noble, Curator of Reptiles at the American Museum of Natural History, asserting that the famous specimen of *Alytes* was faked.

Noble was thirty-two, a graduate of Harvard and Columbia, who described himself in the American *Who's Who* as a 'Curator and Explorer'. He had led several naturalist expeditions to Guadeloupe, Newfoundland, etc. Gregory Bateson describes him as a ruffian, and among English zoologists who knew him he had the same reputation. But he was without doubt an expert on reptiles.



- “scandal”, “faked”, August 1926 - that's just a month before Kammerer's suicide - ouch!
- So why do we talk about reptiles now?

midwife toad



- (German: Geburtshelferkröte)
- mates on land
- nuptial pads (German: Brunftschwien): swellings on the male's palm
- midwife toads **don't have them** as the female's skin is dry on land

The greatest sensation among all of Kammerer's experiments was created by his work with that small ungainly creature, the midwife toad, *Alytes obstetricans*.

While most other toads, and frogs, mate in the water, *Alytes* mates on land. During the mating procedure in water, the male toad clasps the female round the waist and keeps her in his grasp for a considerable time—sometimes for weeks—until she spawns her eggs, which he then fertilises with his sperm. To get a firm grip on the female's slippery body in the water, the male toad develops in the mating season swellings on its palm and fingers of a blackish colour, from which small horny spines protrude: these are the famous nuptial pads. But the midwife toad, mating on land, where the female's skin is comparatively dry and rough, does not need and does not possess these pads. The female emits a multitude of eggs attached to long strings of jelly, and the male, after having fertilised them, winds these beads of eggs round his hind-legs and carries them with him until the young hatch—hence the name midwife toad.*

We still miss the evolutionary part?

Kammerer's claim was that by inducing *Alytes* to copulate in water, like other toads, for several generations, they eventually developed nuptial pads as an acquired hereditary feature.

- enforced change of the environment induces an **adaptation** of the organism over several generations
- sounds like **evolution** - where is the problem?



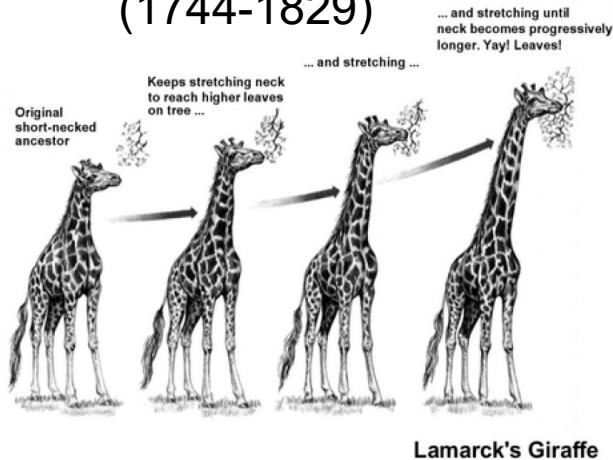
Nevertheless, the hereditary fixation of an acquired, adaptive—or re-adaptive—feature was a puzzling phenomenon, and seemed difficult to reconcile with the ‘no interference with the germ-track’ theory.

- Can “organisms pass to their offspring characteristics **acquired in their lifetime**”?

fight 1: Lamarck vs Darwin



Jean-Baptiste Lamarck
(1744-1829)

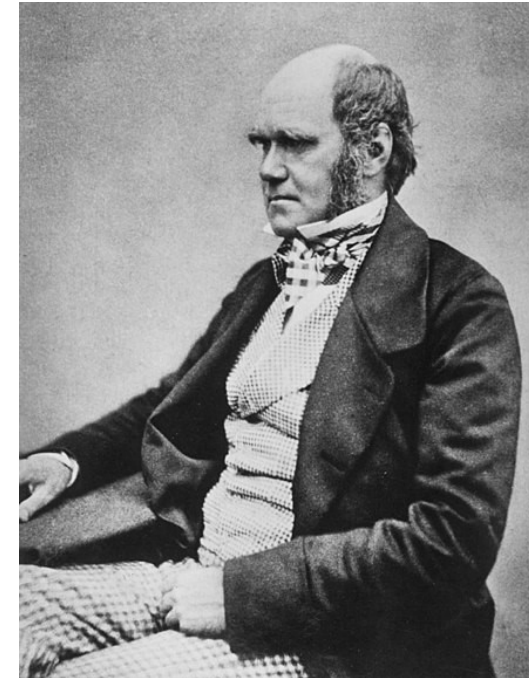


Lamarckism

- inheritance of acquired characteristics
- How would that **information flow from behavior to genes** be implemented?

Darwinism

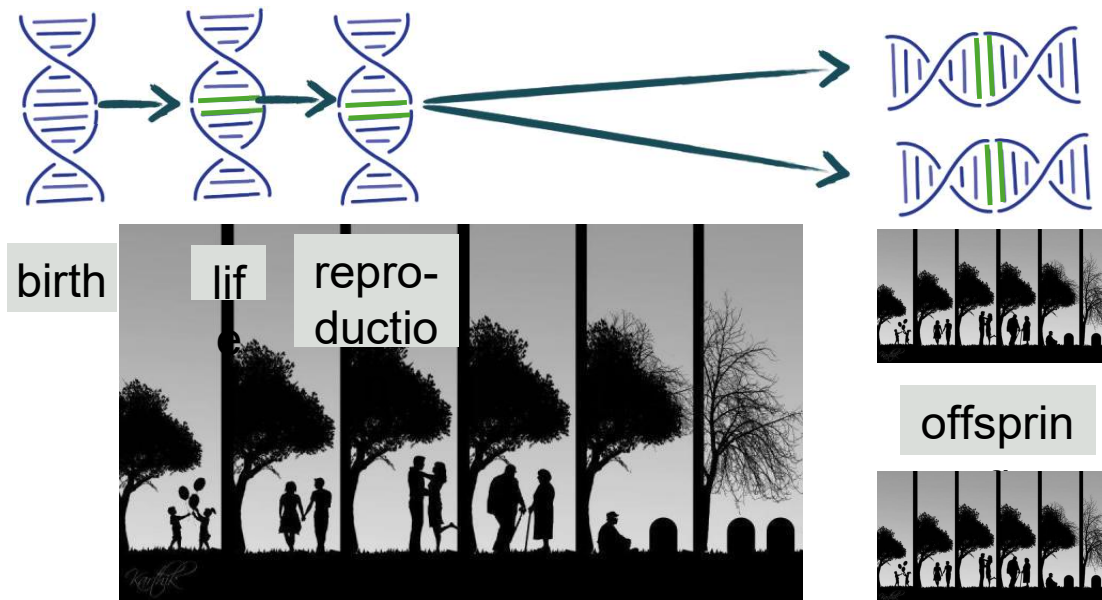
- “natural selection of small, inherited variations that increase the individual’s ability to compete, survive, and reproduce”
- Your genes are determined before birth and are **not changed** in a directed way **during lifetime**.



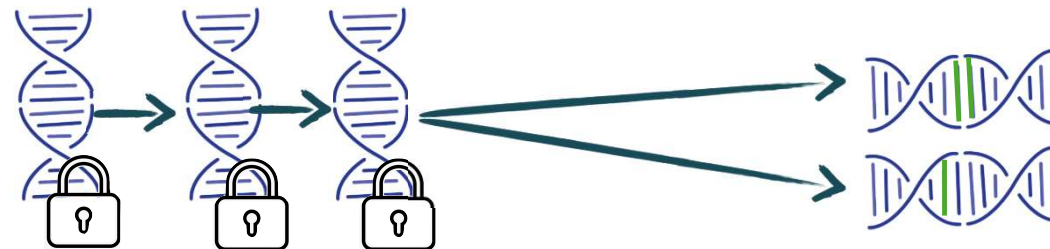
Charles Darwin
(1809-1882)

Lamarckism and Darwinism simplified

Lamarckism



Darwinism

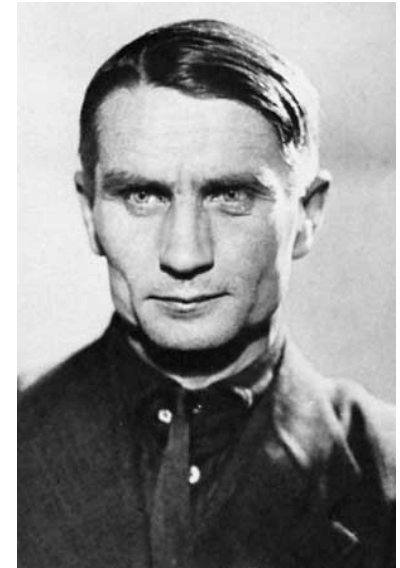


Influence on the Soviet Union

Soviet silent movie, 1928
“Salamandra”
about Kammerer



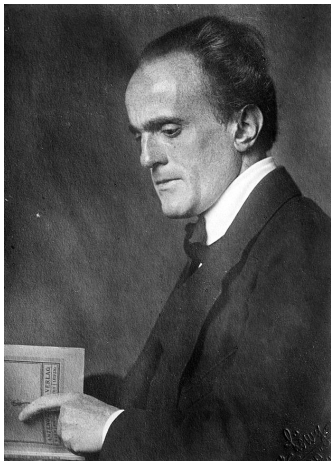
- “A form of Lamarckism was revived in the Soviet Union of the 1930s when Trofim Lysenko promoted the ideologically-driven research program, Lysenkoism; this suited the ideological opposition of Joseph Stalin to genetics. Lysenkoism influenced Soviet agricultural policy which in turn was later blamed for crop failures.”
- he thought Mendel’s theory was too reactionary or idealist (later more on Mendel)



Trofim Lysenko
(1898-1976)

- <https://en.wikipedia.org/wiki/Lamarckism>

fight 2: Kammerer vs Bateson



Paul Kammerer
(1880-1926)

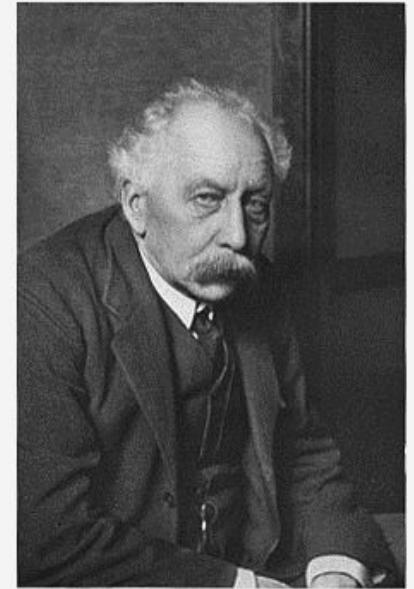
Kammerer

- experiments with midwife toads but also fire salamanders etc. that indicate Lamarckism is real

Bateson

- Bateson had started as Lamarckian, did an unsuccessful expedition to Central Asia, then switched to Mendelism
- he didn't accept Kammerer's results as evidence for Lamarckism, accusations of fraud

Bateson denied the validity of Kammerer's breeding experiments with the midwife toad, but Bateson never succeeded in breeding midwife toads at all.*



W. Bateson

William Bateson
(1861-1926)

The question remains, what is the real nature of the swellings in the animal exhibited? That on the palm did not look like a nuptial pad. What there may have been on the back of the hand I do not know. I made no statement about it, though Dr. Kammerer says I did. I might no doubt have asked to see the back, but I had no reason to suppose there was anything more to see.¹¹

(William Bateson)

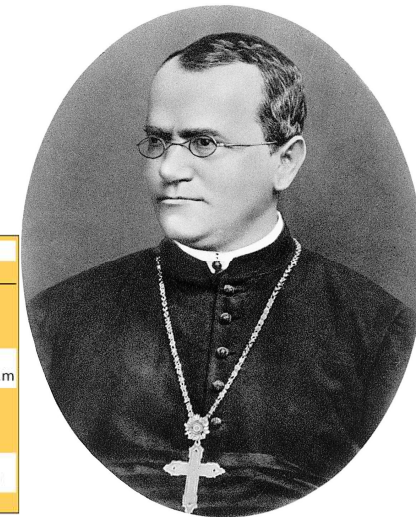
Mendelism / Mendelian inheritance

- Characters are unitary. That is, they are **discrete** (purple vs white, tall vs dwarf).
- Genetic characteristics have alternate forms, each inherited from one of two parents. Today, we call these **alleles**.
- One allele is **dominant** over the other. The phenotype reflects the dominant allele.
- Gametes are created by **random segregation**. Heterozygotic individuals produce gametes with an equal frequency of the two alleles.
- Different traits have independent assortment. In modern terms, **genes are unlinked**.

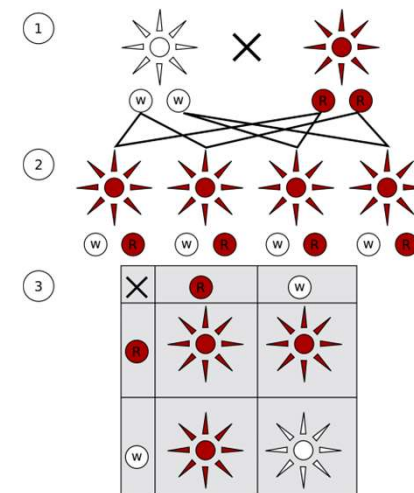
gamete: haploid cell that fuses with haploid cell during fertilization

heterozygotic: 2 different alleles in one gene locus

Samen		Blüte	Schote		Stängel	
Form	Keimblatt	Bräue	Form	Bräue	Ort	Größe
						
grau & rund	gelb	weiß	voll	gelb	mittig Sch., und Blüten	lang 1,8m-2,1m
						
weiß & schrumpelig	grün	violett	verengt	grün	abschl. Sch. Blüten oben	kurz (<30cm)
1	2	3	4	5	6	7



Gregor Mendel
(1822-1884)



dominant and recessive phenotypes

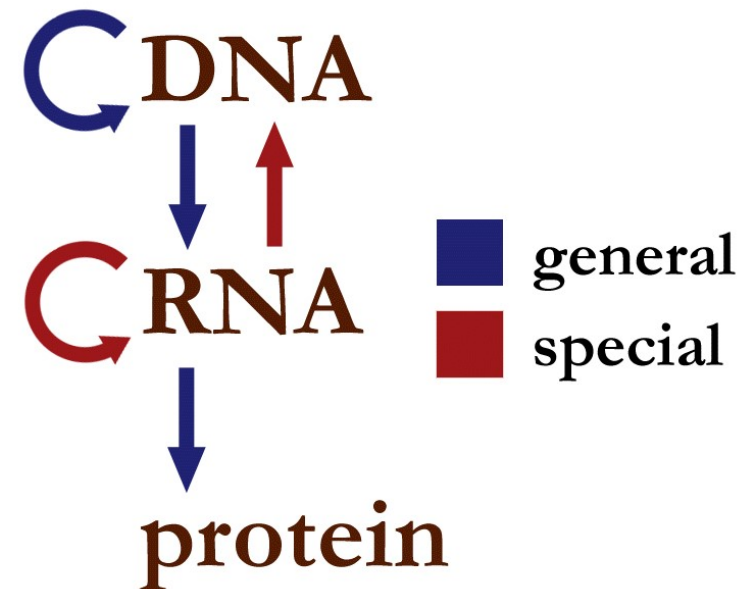
https://en.wikipedia.org/wiki/Mendelian_inheritance

• https://en.wikipedia.org/wiki/Gregor_Mendel

Central Dogma

Then, in the 1950s, came another dawn: the discovery by Crick and Watson of the chemical structure of DNA, the nucleic acid in the chromosomes, carrier of the 'hereditary blueprint'. The Weismann doctrine, that nothing* that happens to an organism in its lifetime can alter that blueprint, was now elevated into Crick's so-called 'Central Dogma', which states that 'information can flow from nucleic acids to proteins [i.e. from blueprint to building block] but cannot flow from protein to nucleic acid'. But dogmas are brittle structures. On June 25, 1970, the *New Scientist* announced: 'Biology's Central Dogma Turned Topsyturvy.' *The Times* Science Report followed suit: 'Big Reverse for Dogma of Biology.' The report concluded:

It is too early to say what consequences may follow from the demonstration that DNA can be copied from RNA,† but at the least the central dogma now seems to be an oversimplification.⁶



Mendel hassle



- Mendelian inheritance makes **predictions** about ratios of finding, e.g. yellow and green beans. Say, the prediction is 3:1.
- Mendel's own **published results** get extremely close to that 3:1.
- too close:
Sir Ronald Fisher calculated that repeating Mendel's experiments would give you such an exact result **only 1 in 2000** experiments.

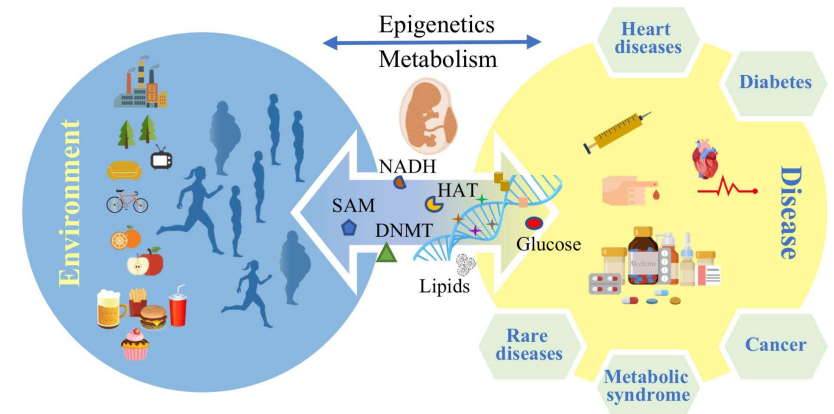
[Fisher, Ronald A. "Has Mendel's work been rediscovered?" *Annals of science* 1.2 (1936): 115-137]



- Mendel was **re-discovered by accident**
- he published in 1865 and was ignored for **35 years** until 1900
- **Wilhelm Focke** published a book *Die Pflanzen-Mischlinge* in 1881 that contains a short note about Mendel's experiments
- a biologist from Oxford wrote an entry for the *Encyclopedia Britannica* and asked **Charles Darwin** for a list of related publications
- Darwin was lazy and just sent him *Die Pflanzen-Mischlinge* with a note that he hadn't read it yet but that it had a nice bibliography

A long story short

- Kammerer's experiments have never been repeated to the day
- He was a master of breeding amphibians and reptiles. Nobody achieved similar successes.
- He had worked at the Viennese Vivarium for years. He lost most of his animals during world war I.
- Kammerer kept samples of toads for display but it was difficult to preserve them.
- One last of them was most likely doctored with ink but probably not by himself (!).
- His suicide was probably not the effect of one cause but multiple (financial problems, relationships etc.).
- His results could possibly be explained today by epigenetic effects but that's also controversial.



Did we learn something for robotics??

- We do artificial evolution and can choose what to implement.
- Do we want to try implementing Lamarckism?
 - whatever you learn at lifetime is transferred to your offspring
- Do we want to try implementing Darwinism?
 - what happens during lifetime is lost
- Could we do both?
- Can we evolve a capability to learn?

